

Exercise - 04

Setting Up a Local Area Network (LAN)

Aim :

To design, set up, and configure a Local Area Network (LAN) connecting four computers through a network switch, assign static IP address to all devices, verify connectivity using the ping command and demonstrate resource/data sharing across the network.

Equipment Required :

S.NO	Equipment	Quantity	Specification
1.	Personal Computer (PC)	4	Ethernet (NIC) enabled
2.	Network Switch	1	8 / 16 / 24 port, unmanaged / managed
3.	Ethernet cables (Cat5e / Cat6)	4	Straight-through, RJ-45 connectors.

Theory :

A Local Area Network (LAN) is a network that connects computers and devices within a limited geographical area such as home, office, or laboratory, allowing them to share resources such as files, printers, and internet connections.

Network Diagram.

The LAN was implemented using a star topology as shown below, with 4 PCs connected individually to a central switch.

Implementation in Cisco Packet Tracer.

Since physical hardware was limited, the LAN was also implemented and verified using Cisco Packet Tracer simulation software, following the steps below.

Step 1 : Place the Devices

1. Open Cisco Packet Tracer
2. From the device category panel, select End Devices and drag 4 PCs onto the workspace.
3. Select switches and drag 1 switch onto the workspace.

Step 2 : Cable the devices

4. Click Connections → Select Copper Straight-Through cable.
5. Click each PC, then click the switch, and select an available Fast Ethernet port to complete each connection.

6. Repeat for all four PCs. Green link lights confirm a successful physical connection.

Step 3 : Assign IP Address to PCs

7. Click a PC → Desktop tab → IP configuration
8. Enter the static IP address and subnet mask per the IP table (section 5) for each PC.

Step 4: configure the switch Management IP

9. click the switch → CLI tab
10. Enter the following commands to assign a management IP
enable
configure terminal
interface Vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit.

Step 5: Verify Connectivity using ping

11. Click a PC → Desktop tab → Command prompt
12. use the ping command to test connectivity to the other PCs
13. Successful "Reply from...." message confirm Connectivity.

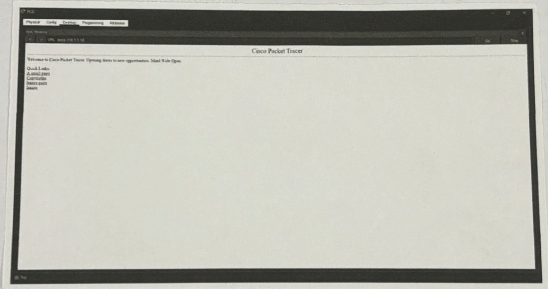
Step 6: Demonstrate Data Sharing (Server Simulation)

14. Add a Server device (End Devices → Server-01) and connect it to the switch.
15. Assign it an IP address (eg, 10.1.1.10, subnet mask 255.0.0.0) via its Desktop → IP Configuration tab.
16. Open the server's (config) tab and enable the HTTP service.
17. From another PC, open the web browser (Desktop tab) and enter the server's IP address to load the page, confirming successful data transfer across the LAN.

Step 7 : Observe packet flow in Simulation Mode.

18. Switch to Simulation mode (Bottom-right of the workshop).

19. Send a ping or web request between two devices and use the play / capture controls to step through the packet as it travels hop-by-hop through the switch, observing MAC - address - based forwarding.



Result :

Thus the exercise is done successfully.